Choosing 2.4 GHz or 5 GHz Wi-Fi

Article: NF-EQ-003 | **Owner:** Network Equipment
Category: equipment | **Topic:** wifi_coverage_bands
Audience: customer_assistant | **Applies to:** Fictional dual-band NF-RTR-200 and compatible devices
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Related articles: NF-TS-003, NF-TS-005, NF-EQ-002

1. Understand the band trade-off

The NF-RTR-200 offers 2.4 GHz and 5 GHz Wi-Fi. The 2.4 GHz band generally reaches farther and works with many older devices, but has less available capacity and more common interference. The 5 GHz band often provides higher throughput near the router but loses strength more quickly through walls. Neither band provides a guaranteed speed at every location. 5 GHz Wi-Fi is not the same as 5G mobile service.

2. Select the band for the task

Try 5 GHz for a compatible computer, television, or phone near the router when throughput matters. Try 2.4 GHz for a distant room or a device that supports only 2.4 GHz, such as some smart-home devices. If combined-band naming is enabled, the router may select a band automatically. A device changing bands is not necessarily a fault. Do not split network names or modify advanced settings unless the customer understands how to reconnect devices and restore the original configuration.

3. Test coverage fairly

Compare the same device near the router and at the problem location using the same application. Note the band, distance, walls, and whether other household devices are busy. If only the distant room performs poorly but wired service is normal, investigate coverage rather than assuming an optical line fault. Devices with older Wi-Fi hardware may limit performance. A high subscribed plan does not upgrade a device's wireless radio or remove structural obstacles.

4. Improve placement before buying equipment

Keep the router central, raised, and out in the open. Avoid metal enclosures, large appliances, and tightly packed cable/electronics areas. For a stationary computer or television, a safely routed Ethernet cable can provide a more stable connection. Keep walkways clear and do not move, pull, or bend the optical fiber to relocate the router. If placement requires changing the installed fiber route, seek an authorized technician rather than extending optical cabling yourself.

5. Extenders, mesh, and compatibility

A third-party extender or mesh system may help coverage, but performance depends on its placement, backhaul, and compatibility. NovaFiber does not promise that buying one will fix a line fault. Optional equipment pricing is not established by this article; request a quote if provider equipment is offered. Do not buy a faster plan solely to solve a dead zone. Verify line performance with NF-TS-003 and discuss the layout with support before choosing additional hardware.

6. When the problem needs another article

If Wi-Fi is connected but websites fail on all devices, use NF-TS-005. If a wired device also performs poorly, use NF-TS-003. Red LOS or abnormal PON requires NF-TS-002 regardless of Wi-Fi strength. If Wi-Fi disappears repeatedly while wired access stays online, note the timing and contact support using NF-TS-004. Nova cannot scan the home, measure radio interference, or guarantee coverage remotely; it can explain the approved comparison checks.